



Air University, Islamabad

Project Proposal

for

Project Title

EZSell – a Marketplace for Buying & Selling Pre-Owned Items

Version 1.0

By

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|---|--|---|
| ☐ A-Desktop Application/Information System | ☒ B-Web Application/Web Application based Information System | |
| ☒ C-Problem Solving and Artificial Intelligence | ☐ D-Simulation and Modeling | ☐ E-Smartphone Application |
| ☐ F-Smartphone Game | ☐ G-Networks | ☐ H-Image Processing |
| ☐ Other (specify category) _____ | | |
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Abstract

EZSell is designed to integrate an AI-driven, straightforward, and trustworthy marketplace for purchasing and selling pre-owned furniture and electronics. Existing platforms have gaps in transparency, resulting in many buyers and sellers being victims of fraudulent pricing. For buyers, understanding the condition of electronics prior to making a purchase can be vastly difficult.

Our project aims to include price prediction algorithms, secure user authentication, and easy-to-use product listing interfaces for enhanced security and transparency. Unlike conventional listing websites, EZSell provides a closed circuit that protects its users and improves their overall experience.

Guaranteeing reasonable pricing is one of our foremost objectives. Having a more streamlined process of buying and selling is also needed. Moreover, users will be able to utilize basic augmented reality features to see how furniture will fit in their space before buying. EZSell strives to reduce the hurdles associated with buying secondhand products to build a more user responsive and friendly marketplace.

1. Introduction

This document outlines the project proposal for the EZSell development, which is a self-learning online marketplace that aims to ease the selling and buying processes of secondhand electronics and furniture. In the current world economy, businesses and consumers are linked and can transact without geographical boundaries, thanks to e-commerce platforms. However, effective online marketplaces are plagued with numerous problems, such as trust issues, unreliable pricing schemes, and even scams, which make it impossible for people to make the best choices. Invalid pricing based on algorithmic prediction and weak verification of identities is what brought EZSell into existence. Artificial intelligence will be used to estimate fair market prices for used items, and secure authentication mechanisms will be adopted to enhance user trust. EZSell hopes users will appreciate the ease of use of the platform in listing items for sale and purchasing items because of the transparency and security availability. Unlike conventional classified ads websites, EZSell will estimate pricing through powerful algorithms, which would significantly reduce losses associated with excessive or insufficient pricing. Furthermore, the application will suffer from countermeasures, such as abnormal user account registration for the purpose of listing fake items and scamming people. As a website application, it would be possible to access the solution using all kinds of devices, which would improve the user experience perfectly. The project is aimed at building a reliable secondhand marketplace by concentrating on user engagement, market efficiency, and approaches to pricing that are considered ethical.

2. Problem Statement

Pakistan's used electronics and furniture marketplace confronts a few serious problems, such as untrustworthy pricing schemes, deceitful ads, and insufficient buyer protection. Several online classified advertisements function as auction sites where sellers price their goods at will. They often suffer from posting price tags that defy rational economic logic. In the absence of a comprehensive and universal pricing system, it is a very difficult and challenging task for the users to tell the market value of the item. Furthermore, fake accounts combined with misleading descriptions of products greatly contribute to the distrust of users, which discourages them from participating. These systems also lack the AI-powered traditional system analysis features that ameliorate these shortcomings. A clearly defined intelligent marketplace is needed to provide users with the needed confidence and trust.

3. Problem Solution

In order to resolve issues in the existing secondhand marketplace, EZSell has been developed as an AI-powered system that guarantees appropriate user pricing alongside enhanced trust. Unlike traditional classified advertisement systems, EZSell determined intelligent pricing models will recommend better-automated pricing to users, which will artificially reduce over- and underpricing. There will be a strong user verification system in order to build trust among buyers and sellers and eradicate fraudulent listings.

Smart search filters will be available to all users, which will provide efficient searching options based on specifications, condition, and its range. EZSell's goal is to revolutionize secondhand marketing systems by making them easy to be sold.

3.1 Objectives

- BO-1: Initialising/adding AI-assisted pricing for accurate pricing suggestions for secondhand electronics and furniture.
- BO-2: Strengthen user verification and fraud detection systems to decrease fraudulent listings by 70%.
- BO-3: Implement filters for brand, model, condition, and price range to enable intelligent search functions.
- BO-4: Enable secure communication between buyers and sellers.
- BO-5: Flag misleading product descriptions and detect suspicious behavior.
- BO-6: Enable relevant product suggestions based on user activity and search history.
- BO-7: Revamp the marketplace from a classified ad system to an organized e-commerce platform for secondhand goods.
- BO-8: Implement an efficient ad posting interface to streamline the listing process by 30%.

4. Related System Analysis

Following is the overview of three such systems.

1. OLX

OLX is one of the platforms commonly used for trading of used phones, laptops, and other electronics. However, it has no AI pricing system, which leads to problems in estimating the prices of items. Moreover, fraudulent advertisements remain a significant problem because of insufficient means of verification.

2. eBay

Users can dispose of both brand new and secondhand items in an auction or direct sale format on eBay. It is well known among buyers, however, that the system has a long and challenging bidding process along with high transaction costs for less frequent sellers. eBay does not concentrate on local sales, which renders it ineffective for rapid consumer-to-consumer sales.

3. Facebook

Through Facebook, users can engage in buying and selling between members of their local vicinity. An advantage offered as verification through social networks provides is that users' profiles allow an account holder to verify their identity. Also, there is no automatic price set as well as no respective classes for products.

Application Name	Weakness	Proposed Project Solution
OLX	No AI-driven pricing; high risk of scams.	AI-based price prediction; improved user verification.
eBay	High listing fees; complex auction process.	Affordable listing fees; instant price suggestions for faster sales.
Facebook Marketplace	Lack of structured pricing; scam risks.	AI-powered fair pricing; seller credibility scoring.

5. Vision Statement

For people and companies wishing to trade used mobile phones, laptops, or tablets at their own convenience but having obstacles in arriving at a fair price and having their items sold without being scammed, The EZSell Marketplace, an AI-powered online platform, integrates technologies to offer accurate price predictions, user verification, and automated selling. Contrary to existing platforms like OLX and Facebook Marketplace, which have no AI-integrated pricing, have high chances of fraud, and give users low satisfaction, our product helps users by giving the appropriate pricing, increasing security through verified transactions, and improving user experience by optimizing with AI.

6. Scope

EZSell is a mobile and online marketplace powered by AI that makes selling and purchasing used mobile devices, laptops, tablets, and furniture safe and efficient. The platform will improve user experience by leveraging AI algorithms to offer predictive analysis of prices, ensuring that users will be able to price their products realistically. For safety, user verification mechanisms will be included in order to reduce the chances of any negative terms of usability. The marketplace will allow users to easily provide product information, images, and details through a simple product posting interface. Using historical sales data and market trends, prices will be set automatically by a recommendation engine. Integrated messaging will also be provided to enable secure communications between sellers and buyers. Trust will be built in the marketplace community by implementing a ratings and review system. Market information will be gathered for price evaluation and comparison through web scraping. In order to enhance Users interactions, they will be able to access the marketplace through web applications over the web, which will make it easier for everyone to use. An admin dashboard will incorporate user and policy management features. The system will also include search and filter options for users to find the most suitable product and the product they want for themselves.

7. Modules

7.1. Profile Management Module

- FE-1: Verify user identity and enable registration.
- FE-2: Allow users to securely reset passwords.
-

7.2. Product Listings and Management Module

- FE-1: allows users to create ads for used mobile phones, laptops, tablets, and furniture, including images and descriptions.
- FE-2: Users can edit, update, or remove product listings as desired.
- FE-3: view listing analytics, including impressions and interest from possible buyers.

7.3. Price Prediction Module

- FE-1: Provides real-time price prediction based on the market trend using AI.
- FE-2: Price suggestions for sellers to help in setting up prices competitively.

7.4. Prices and Product Recommendations Module

- FE-1: Search filters allow perusal according to brand, price, condition, and location.
- FE-2: Recommendations for products based on the user's previous activities on the site.
- FE-3: Save listings to favorites and get notifications whenever there is a change in sale price.

7.5. Analytical Dashboard Module

- FE-1: Detecting seasonal and regional price differences.
- FE-2: Give information on the brands and products known to be highly searched for
- .

7.6. User Engagement Analytics Module

- FE-1: Check the interactions and engagement of users per listing.
- FE-2: Give attention to the brands and products the consumers are always searching for.
- FE-3: Based on the user's browsing and purchase history, notify the user personally.

7.7. Try-On & Customization Module

- FE-1: Users can customize furniture by uploading their chosen images along with their room's background decor.
- FE-2: Users are able to place furniture on the image of the room and calibrate their positioning.
- FE-3: Users can reposition furniture as they desire in order to best visualize it in the context of their room.

7.8. Fraud Prevention & Secure Listings Module

- FE-1: Users are verified by cross-checking their provided phone numbers.
- FE-2: Moderation of user-generated content by admin prior to acceptance.
- FE-3: Blocking or deletion of ads that infringe policies without manual review.

8. System Limitations/Constraints

- LI-1: The system will support only web-based access on business launch. There will be mobile responsiveness, but no dedicated mobile app.
- LI-2: At launch, AI-based price prediction features will focus only on electronic items such as mobiles, tablets and laptops. There are expansion plans for other product categories later.
- LI-3: In the first phase, user verification will be based on third-party verification only (email and phone verification) and will not include biometric verification.
- LI-4: Web scraping for price comparison will have to rely on publicly available data, which can be modified by the websites in real time, thus limiting their accuracy.
- LI-5: The platform will allow communication between buyers and sellers but not the collection of payments through the website during the first release.

9. Data Gathering Approach

In order to meet user demands as well as marketplace requirements, a multimethod approach to data collection is employed within the system:

- Interviews: Engaging with prospective users (buying and selling) of the marketplace to gauge feedback and expectations related to the proposed marketplace.
- Competition Study: Review known players in the field, such as OLX, eBay, or Facebook Marketplace, and their shortcomings for further development analysis.
- Scraping the Web & Market Surveys: Analyzing available information on the internet regarding the changes in the values of products to enhance the machine learning-based estimation of price prediction.

10. Tools and Technologies

Table 1: Tools and Technologies

Tools & Technologies	Version	Rationale
Programming Languages		
Python	3.11	Applied in constructing AI-driven Price prediction systems and backend infrastructure
JavaScript	ES6+	Primary scripting for client-side development
HTML & CSS	Latest	Design and format the web interface
Frontend Framework		
React.js	18	Development of the interactive dynamic user interface
Bootstrap	5	Styles and layout-based responsive design framework
Backend Framework		
Django	4.2	Backend development web framework responsible for authentication and managing the database
Django REST Framework	Latest	Creation of the interface that connects the frontend and backend systems
Database		
PostgreSQL	15	Used in storing data for web application
APIs & Libraries		
NumPy	Latest	Cleaning data for AI models
Pandas	Latest	Working with big data for price prediction
BeautifulSoup/Scrapy	Latest	Obtaining market data through web scraping
Firebase Authentication	Latest	User login and validation
Development & Collaboration Tools		
Visual Studio Code	Latest	Code editor for frontend and backend development
Git & GitHub	Latest	Version control
Postman	Latest	API testing and debugging
Cloud & Deployment		
AWS EC2 / Heroku	Latest	Hosting and deployment of the web application
AWS S3	Latest	Storage for images and user uploads

11. Project Stakeholders and Roles

Table 2 Project Stakeholders for Proposed Project

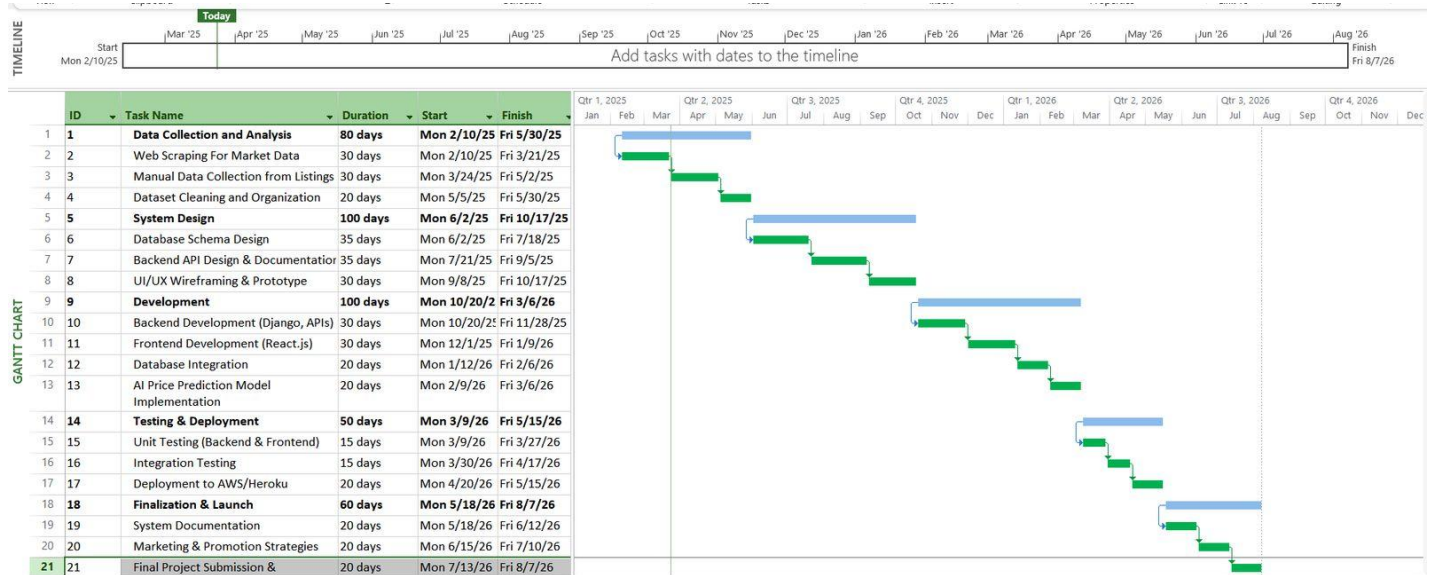
Stakeholder	Role & Responsibilities
Project Sponsor	AIR University Islamabad, Islamabad Campus
Students	Muhammad Ahmed, Ahmed Abbasi, Dawood Haider – In charge of the project construction and its execution
Project Supervisor	Miss Madiha – She assists in the guidance and supervision of the technical and operational components of the project.
Final Year Project Committee	Assesses the project at various phases and suggests improvements.

12. Module-based Work Division

Table 3 Team Member Work Division for Proposed Project

Student Name	Registration Number	Modules
Muhammad Ahmed	222550	Backend Development: - 7.1 Profile Management Module (User authentication, account management) - 7.2 Product Listings and Management Module (Backend logic, ad posting, data handling) - 7.3 Price Prediction Module (AI model training, price calculations)
Ahmed Abbasi	220965	Frontend Development: - 7.1 Profile Management Module (User authentication UI, profile setup) - 7.2 Product Listings and Management Module (Ad posting forms, product display UI) - 7.4 Prices and Product Recommendations Module (Filters, AI-powered recommendations UI) - 7.7 Try-On & Customization Module (Frontend for furniture placement UI)
Dawood Haider	221574	Data Collection, Testing & Deployment: - 7.5 Analytical Dashboard Module (Market analysis, seller insights) - 7.6 User Engagement Analytics Module (Tracking interactions, analytics reporting) - 7.8 Fraud Prevention & Secure Listings Module (AI-based fraud detection, verification mechanisms) - 7.3 Price Prediction Module (Data collection & training support for AI model)

13. WBS and Gantt Chart



14. Mockups

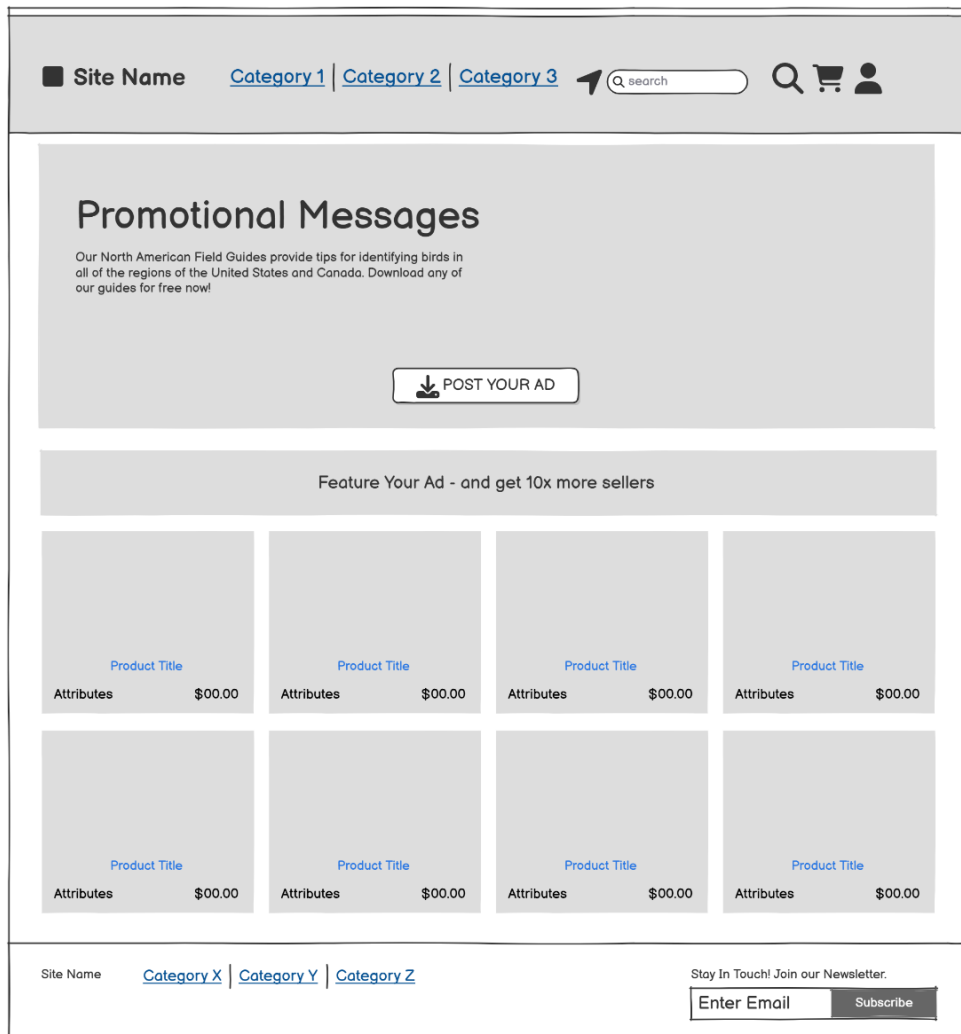


Figure 1: Attractive landing page with promotional deals and placed advertisements to capture and increase user interaction and engagement.



Pakistan

Login

Post Your Ad

Category:
Mobile Phones ▾

Upload Images:
 No file chosen

Brand:
Apple ▾

Ad Title:

Description:

Location:
Islamabad ▾

Price:

Name:

Mobile Phone Number:

Figure 2: A user advertisement page where users can sell mobile phones, tablets, and laptop computers with full specifications.

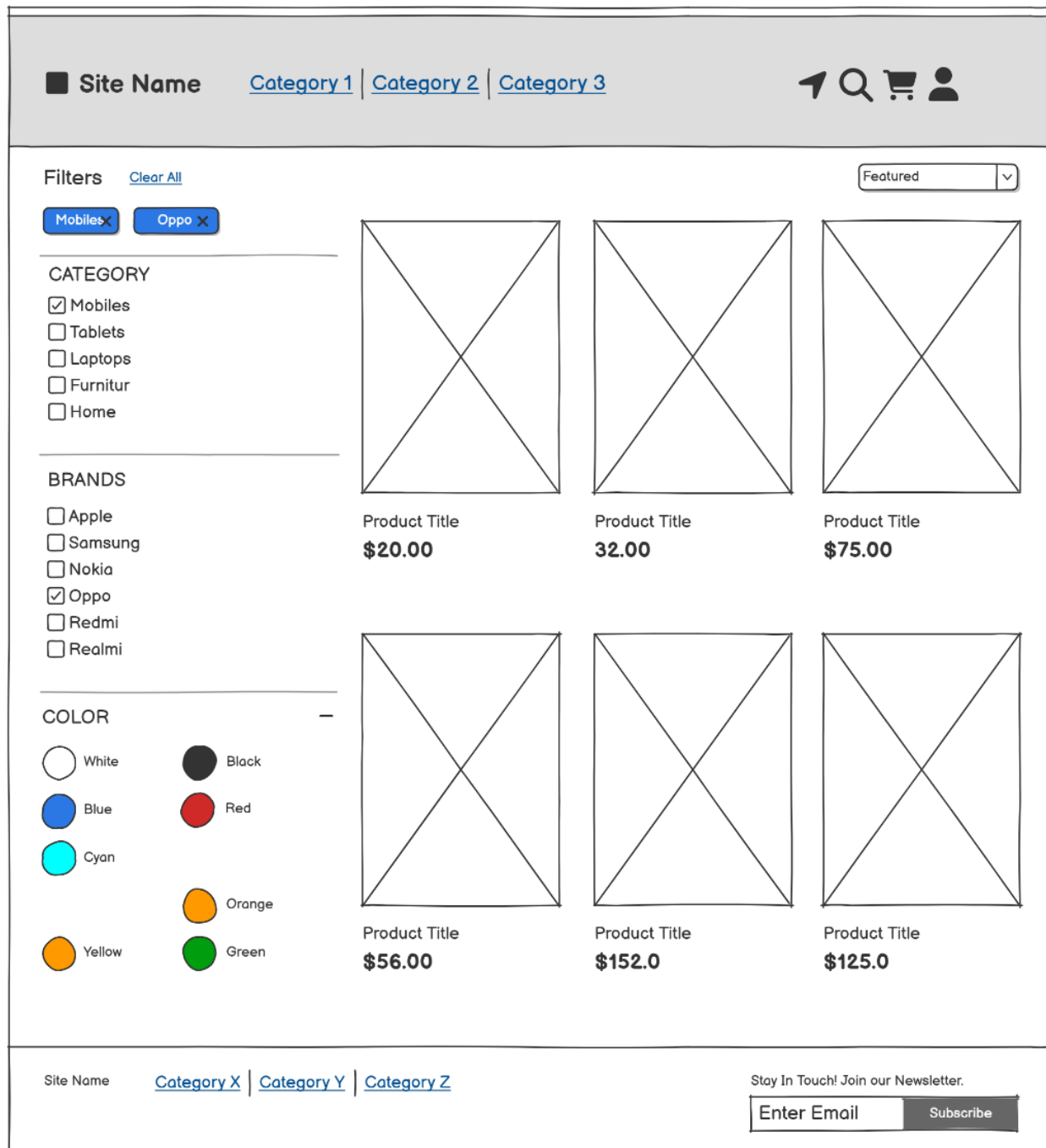


Figure 3: Product search results page that includes advanced filtering possibilities per product category and brand.

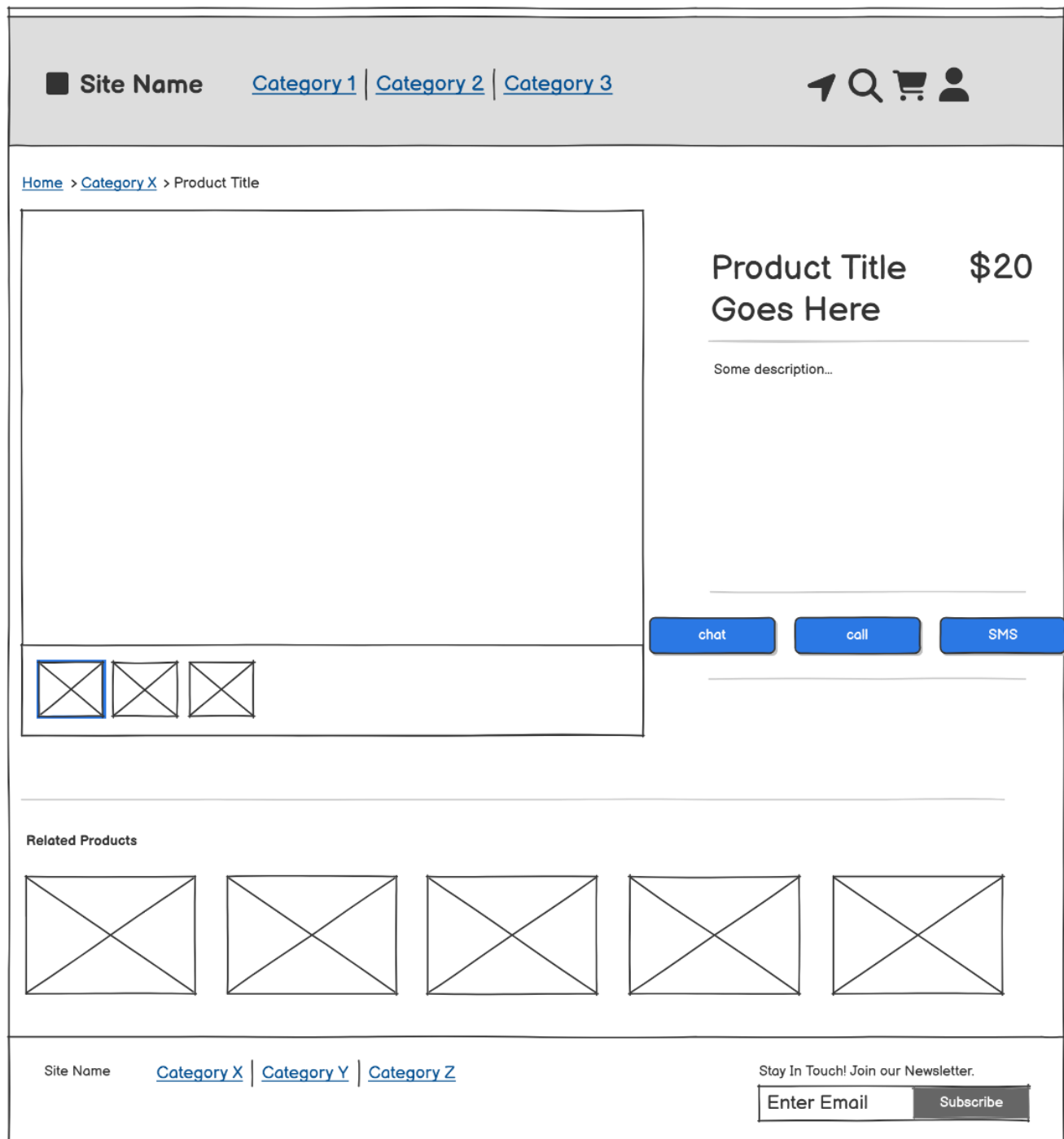


Figure 4: Comprehensive product page containing product image, product description, and communication buttons for buyer and seller to contact each other directly.

Site Name

[Category 1](#)
[Category 2](#)
[Category 3](#)

[Home](#) > [Category X](#) > Customisation

Upload Room Image

Upload Furniture Image

preview

Related Products

Site Name

[Category X](#)
[Category Y](#)
[Category Z](#)

Stay In Touch! Join our Newsletter.

Enter Email

Subscribe

Figure 5: Comprehensive product page containing product image, product description, and communication buttons for buyer and seller to contact each other directly.

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